Single-Phase Boost PFC: Performance and Control

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Abstract

A conventional diode-bridge boost power-factor-correction converter is studied using average current-mode control. Switching-state equations, reference synthesis, duty feasibility, current ripple, and discontinuous conduction are derived. Native SIMPLIS waveforms verify current shaping and identify zero-crossing distortion at 230 V RMS, 380 V DC, 1 kW, and 200 kHz.

I. Introduction

For a sinusoidal source, resistive input behavior requires

$$v_s(t) = V_m \sin\theta, \quad i_s^*(t) = G v_s(t), \quad \theta = \omega t, \qquad (1)$$

where $V_s = V_m/\sqrt{2}$ is the source RMS voltage, $\omega = 2\pi f_{\text{line}}$, and G is the commanded input conductance. Ideally,

$$\text{PF} = \frac{\langle v_s i_s \rangle}{V_s I_s} = 1, \qquad P_{\text{in}} = G V_s^2. \qquad (2)$$

An inner average-current loop tracks a rectified sinusoidal reference; an outer voltage loop adjusts its amplitude to regulate the DC bus [1]. The boost output must exceed the rectified input peak.

A. Circuit Architecture and Scope

The circuit in Fig. 1 comprises a four-diode bridge, L_2, MOSFET Q_1, boost diode D_5, capacitor C_2, and load R_2. Figure 2 reproduces the implemented controller network. A UC3854 average-current macromodel provides the explicit line multiplier and RMS feedforward [5]. Simulations use SIMetrix/SIMPLIS 8.20; supplementary plots are produced in Python from exported simulation data.

TABLE I
Nominal Active-PFC Conditions

Quantity	Value / implementation
Input	230 V RMS, 50 Hz; $R_s = 0.2\,\Omega$
Output	380 V DC; $P_o = 1\,\text{kW}$
Boost stage	$L = L_2 = 300\,\mu\text{H}$; $f_s = 200\,\text{kHz}$
Capacitors	$C = C_2 = 540\,\mu\text{F}$; $C_1 = 330\,\text{nF}$
Parasitics	L_2 ESR and C_2 ESR: 50 mΩ each
Power devices	IRF450 MOSFET; MUR860 boost diode
Load	$R = R_2 = V_o^2/P_o = 144.4\,\Omega$
Current reference	$R_3 = 0.1\,\Omega$ shunt; $R_6 = 4\,\text{k}\Omega$
Voltage compensation	$R_z = 5.6\,\text{k}\Omega$; $C_8 = 4.7\,\mu\text{F}$; $R_{14} = 100\,\text{M}\Omega$
Initialization	Precharged bus; seeded controller states

The ratings follow TI PMP12074 [2], which uses UCC28180 rather than UC3854. This is an architecture/rating benchmark; simulation does not validate TI hardware efficiency.

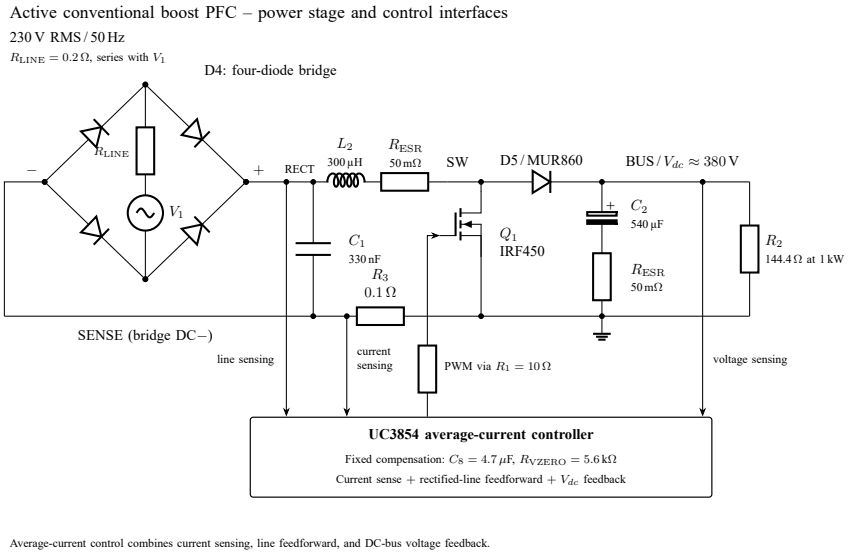


Fig. 1. Conventional bridge–boost PFC architecture and UC3854 interfaces. C_1 is connected across the rectified bridge output; the bulk capacitor C_2 is on the DC bus.

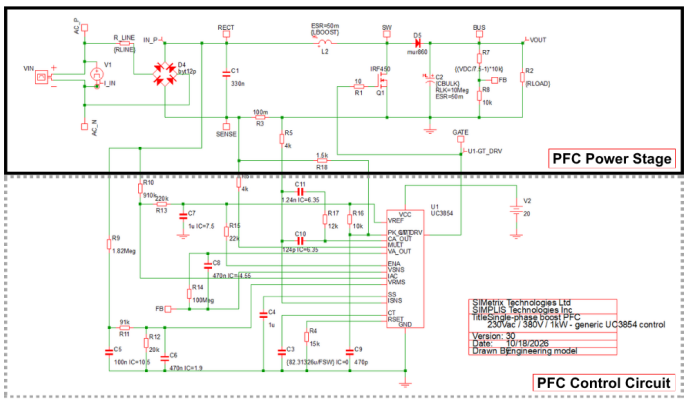


Fig. 2. SIMPLIS implementation of the boost power stage and UC3854 average-current control network.

B. Definitions and Modeling Assumptions

Let v_r denote the voltage at the rectified boost input, v_o the instantaneous bus voltage, $V_o = \langle v_o \rangle$ its DC mean, and i_L the instantaneous inductor current. Define

$$I = \langle i_L \rangle_s , \qquad T_s = 1/f_s, \qquad d = t_{\text{on}}/T_s. \tag{3}$$

Here $\langle \cdot \rangle_s$ is a switching-period average. Ideal assumptions: CCM away from crossings, $f_s \gg f_{\text{line}}$, small bus ripple, and $\min(v_o) > \max(v_r)$. Simulation retains device drops, source resistance, C_1, limits, and local DCM. Use $v_r \simeq |v_s|$ only for ideal estimates.

II. Point 1: Input-Current Shaping

A. Switching States and Averaged Equations

With Q_1 ON, the boost diode blocks and the capacitor supplies the load:

$$\left.\begin{aligned} L\dot{i}_L &= v_r, \\ C\dot{v}_o &= -v_o/R. \end{aligned}\right\} \quad Q_1 \text{ ON.} \tag{4}$$

With Q_1 OFF and $i_L > 0$, the diode transfers inductor energy to the bus:

$$\left.\begin{aligned} L\dot{i}_L &= v_r - v_o, \\ C\dot{v}_o &= i_L - v_o/R. \end{aligned}\right\} \quad Q_1 \text{ OFF.} \tag{5}$$

Weighting (4) and (5) by d and $1-d$ gives

$$\begin{aligned} L\dot{I} &= v_r - (1-d)v_o, \\ C\dot{v}_o &= (1-d)I - v_o/R. \end{aligned} \tag{6}$$

These equations retain line-frequency dynamics while removing switching ripple.

B. Reference Synthesis and Current Tracking

For the ideal bridge, the desired rectified current is

$$I^* = Gv_r \simeq GV_m|\sin\theta|. \tag{7}$$

From (2), with conversion efficiency $\eta = P_o/P_{\text{in}}$,

$$G = \frac{P_{\text{in}}}{V_s^2} = \frac{P_o}{\eta V_s^2}, \quad I_m = \frac{\sqrt{2}P_{\text{in}}}{V_s}, \quad R_{\text{in,eq}} = G^{-1}. \tag{8}$$

The voltage regulator sets G; the multiplier shapes the reference. In the UC3854 macromodel,

$$I_{\text{ref}} = \frac{R_6}{R_3} I_{\text{mult}} = 40000 I_{\text{mult}}. \tag{9}$$

I_{mult} is the actual controller multiplier-output current. It is distinct from an ideal sinusoid and has an approximately $0.32\,\text{A}$ equivalent floor near crossings.

The bridge-output capacitor modifies the relationship between source and inductor currents:

$$\begin{aligned} j_{\text{bridge}} &= i_L + C_1 \frac{\mathrm{d}v_r}{\mathrm{d}t}, \\ i_s &\simeq \operatorname{sgn}(v_s) j_{\text{bridge}}. \end{aligned} \tag{10}$$

The bridge-node relation includes C_1 current, so source current is not simply the signed inductor current. Source PF is measured at the AC source.

Linearizing (6) at a frozen CCM line operating point (D, V_o, I) yields

$$Ls\widehat{I} = \widehat{v}_r - (1-D)\widehat{v}_o + V_o\widehat{d}. \tag{11}$$

Holding $\widehat{v}_r = \widehat{v}_o = 0$, the inner plant and a generic fixed-ramp current loop become

$$\begin{aligned} G_{id}(s) &= \frac{\widehat{I}}{\widehat{d}} \simeq \frac{V_o}{Ls}, \\ T_i(s) &= \frac{C_i(s)G_{id}(s)}{V_{\text{ramp}}}, \qquad \frac{\widehat{I}}{\widehat{I^*}} = \frac{T_i}{1+T_i}. \end{aligned} \tag{12}$$

C_i maps current error to control voltage; V_{ramp} is the PWM ramp amplitude. Large $|T_i|$ gives tracking. This local model does not identify every IC block.

C. Duty Feasibility Near a Zero Crossing

Solving the first averaged equation for the duty required to follow I^* gives

$$d_{\text{req}} = 1 - \frac{v_r}{v_o} + \frac{L}{v_o}\dot{I}^*, \qquad d_{\text{qs}} \simeq 1 - \frac{v_r}{V_o}. \tag{13}$$

For $0 < \theta < \pi$, take $v_r = V_m \sin\theta$ and $v_o \simeq V_o$:

$$d_{\text{req}} = 1 - \frac{V_m}{V_o}\sin\theta + \frac{LGV_m\omega}{V_o}\cos\theta. \tag{14}$$

On the rising quarter-cycle,

$$d_{\mathrm{req}} \le 1 \iff \sin\theta \ge \omega LG\cos\theta \iff \theta \ge \tan^{-1}(\omega LG). \tag{15}$$

At $v_r = 0$, even $d = 1$ gives $\dot{I} = 0$, whereas an ideal sine demands a positive slope. This bound excludes $d_{\max} < 1$, drops, offsets, and C_1; it does not predict the complete distortion width.

D. Switching Ripple and CCM Boundary

The ON-state current rise gives the peak-to-peak switching ripple:

$$\Delta i_L \simeq \frac{v_r d}{L f_s} \simeq \frac{v_r(1 - v_r/V_o)}{L f_s}. \tag{16}$$

Since

$$\frac{\mathrm{d}}{\mathrm{d}v_r}\left[v_r(1 - v_r/V_o)\right] = 1 - \frac{2v_r}{V_o}, \tag{17}$$

the ripple maximum is

$$\Delta i_{L,\max} = \begin{cases} \dfrac{V_o}{4Lf_s}, & V_m \ge V_o/2, \\ \dfrac{V_m(1 - V_m/V_o)}{Lf_s}, & V_m < V_o/2. \end{cases} \tag{18}$$

At the nominal point, $V_m = 325.269\,\mathrm{V}$, so $\Delta i_{L,\max} = 1.583\,\mathrm{A}$ at $v_r = 190\,\mathrm{V}$; the ideal line-crest ripple is $0.7815\,\mathrm{A}$. The largest ripple need not occur at the line crest.

The approximate CCM condition is

$$I_{\mathrm{valley}} \simeq G v_r - \frac{\Delta i_L}{2} > 0. \tag{19}$$

Away from $v_r = 0$, substitution of (16) gives

$$G > \frac{1 - v_r/V_o}{2Lf_s}. \tag{20}$$

Reducing load lowers G and increases the likelihood of local DCM. Nominal CCM does not imply continuous current at every AC crossing.

E. Discontinuous-Conduction Relation

Let δ be the diode-conduction fraction, with idle fraction $1 - d - \delta$. For a cycle starting at zero current,

$$I_{\mathrm{pk}} = \frac{v_r d}{L f_s}, \quad \delta = \frac{v_r d}{V_o - v_r}, \quad d + \delta \le 1. \tag{21}$$

The falling slope is $(v_r - V_o)/L$; equating rise and fall gives δ. The triangular current area then yields

$$I = \frac{I_{\mathrm{pk}}(d + \delta)}{2} = \frac{v_r V_o d^2}{2Lf_s(V_o - v_r)}. \tag{22}$$

Imposing $I = G v_r$ gives

$$d_{\mathrm{DCM}} = \sqrt{2Lf_s G(1 - v_r/V_o)}. \tag{23}$$

Use (23) only when (21) satisfies $d + \delta \le 1$. Both expressions assume ideal devices and nearly constant voltages during T_s.

F. Native SIMPLIS Evidence

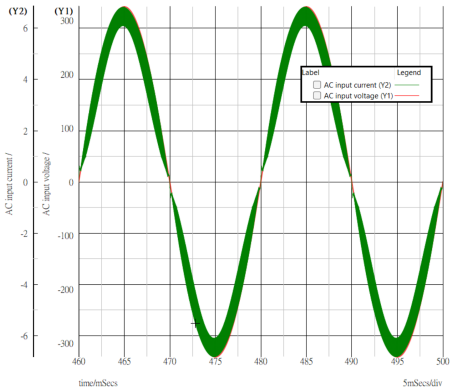


Fig. 3. Native SIMPLIS source voltage and current, 460–500 ms.

Figure 3 shows the source current following the sinusoidal source voltage with switching ripple superimposed. The internal multiplier reference is examined separately in Fig. 4.

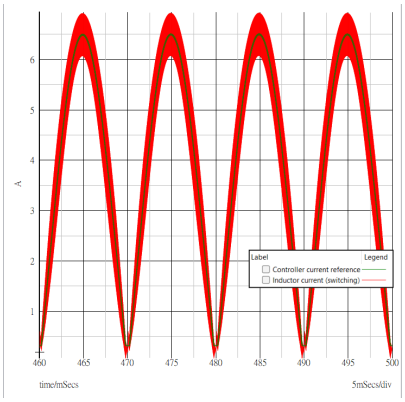


Fig. 4. SIMPLIS inductor current and reference, 460–500 ms. Table II reports the 780–800 ms baseline.

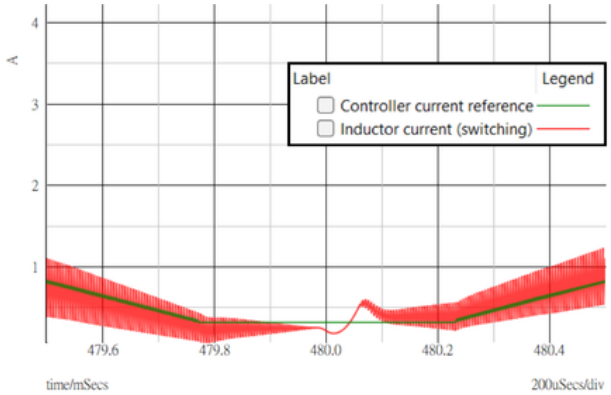


Fig. 5. SIMPLIS crossing detail, 479.5–480.5 ms. The reference floor and current mismatch are visible.

Figures 4 and 5 distinguish reference shaping from perfect sinusoidal tracking. The roughly $0.32\,\mathrm{A}$ reference floor is

specific to this model. Duty feasibility (15), tracking limits, C_1 dynamics, and possible local DCM also contribute; the plots do not isolate a single cause.

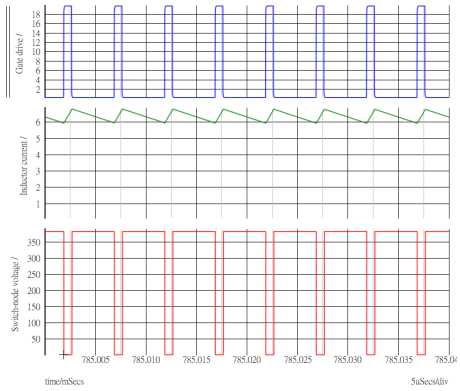


Fig. 6. SIMPLIS waveforms near 785 ms. Top to bottom: gate, inductor current, and switch-node voltage.

In Fig. 6, the gate-high interval coincides with a low switch-node voltage and rising i_L; gate-low operation transfers current through the boost diode and reduces i_L, consistent with (4)–(5).

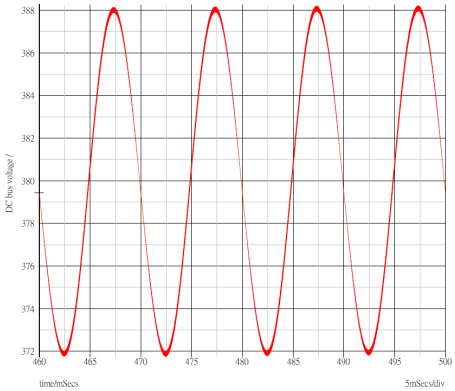


Fig. 7. Native DC bus, 460–500 ms: regulation near 380 V with twice-line ripple. Final-baseline values: Table II.

TABLE II
FULL-LOAD BASELINE: SIMULATION AND THEORY

Quantity	SIMPLIS data	Theory / interpretation
f_s	199.99975 kHz	$T_s \simeq 5\,\mu$s
Mean V_o	380.0575 V	380 V target
Bus ripple	16.1585 V pp	Dominant $2f_{\text{line}} = 100$ Hz
Crest Δi_L	0.8597 A pp	Ideal: 0.7815 A pp
Maximum ripple	Not measured here	Ideal bound: 1.583 A pp
Reference floor	≈ 0.32 A	Macromodel offset/clamp

Steady-state metrics are taken from the measured 780–800 ms baseline and gate-aligned crest-cycle measurements [5]. The measured crest ripple exceeds the ideal value by approximately 10.0%; (16) neglects device drops and actual duty/current dynamics. No isolated loss or error attribution is inferred.

G. Additional Evidence: Export-Based Plots

Figure 8 shows switching-cycle averages and reference overlays over 780–800 ms, with crossing detail centered at 790 ms. Figures 4–5 cover 460–500 ms; tabulated comparisons use the 780–800 ms baseline.

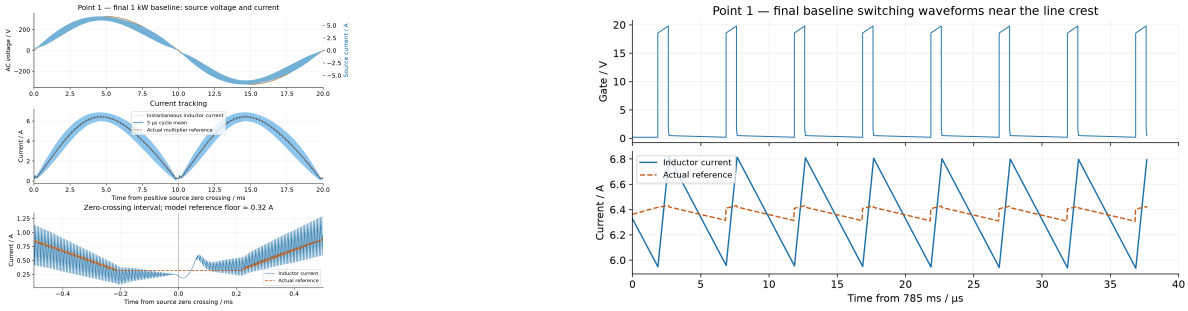
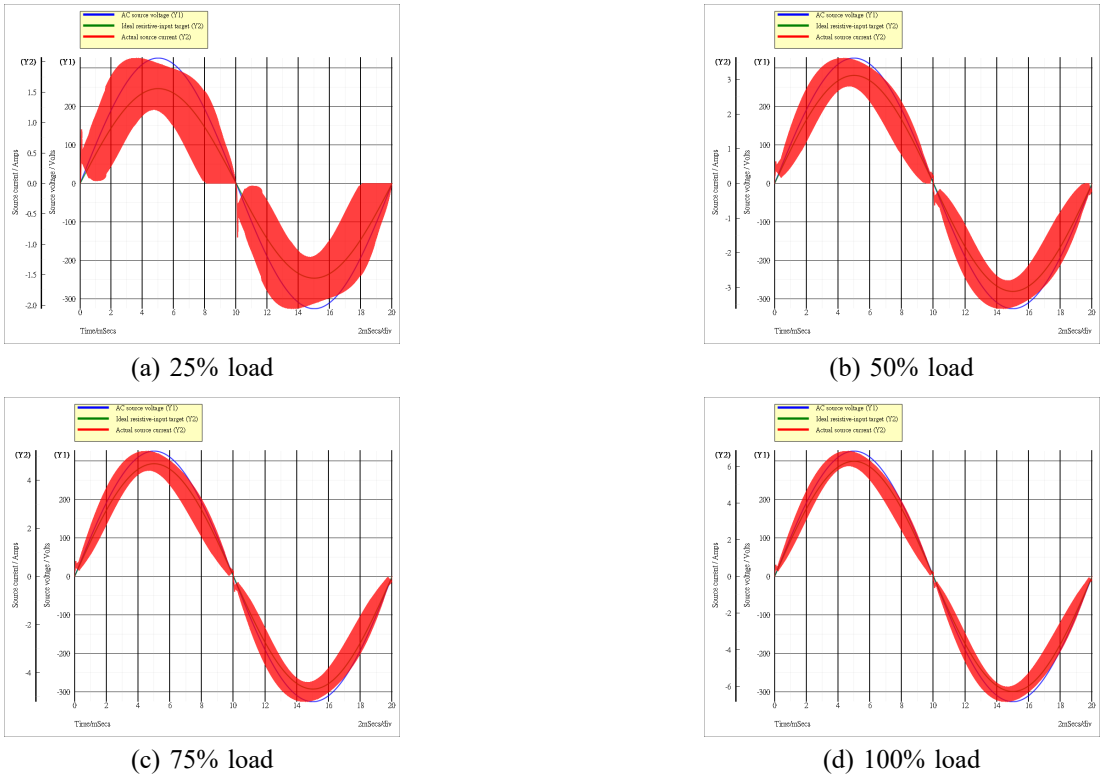


Fig. 8. Additional evidence: postprocessed SIMPLIS data. Left: switching-mean/reference tracking and crossing detail. Right: switching-cycle current tracking at the baseline operating point.

III. Point 2: Power-Factor Analysis



(a) 25% load (b) 50% load

(c) 75% load (d) 100% load

Fig. 9. SIMPLIS load-current waveforms with source voltage and the resistive-input target $(P_{\text{in}}/V_s^2)v_s$. Horizontal origin: positive source crossing in the final line cycle. This ideal target is distinct from the internal multiplier reference.

A. Power and RMS Definitions

For a settled window $T_w = m/f_{\text{line}}$, with integer m,

$$\langle x \rangle = \frac{1}{T_w} \int_{t_0}^{t_0+T_w} x(t)\,\mathrm{d}t. \tag{24}$$

The measured source quantities are

$$V_s = \sqrt{\langle v_s^2 \rangle}, \quad I_s = \sqrt{\langle i_s^2 \rangle}, \quad P_{\text{in}} = \langle v_s i_s \rangle\,, \tag{25}$$

$$\text{PF} = \frac{P_{\text{in}}}{V_s I_s}, \qquad P_o = \frac{\langle v_o^2 \rangle}{R}, \qquad V_o = \langle v_o \rangle\,. \tag{26}$$

Use the AC source current, including represented switching ripple. For a resistive load, $P_o \neq V_o^2/R$ exactly when the bus contains ripple.

B. Displacement and Distortion Factors

Write the periodic source current as

$$i_s = I_{\text{DC}} + \sqrt{2}\sum_{n=1}^{\infty} I_n \sin(n\omega t - \phi_n). \tag{27}$$

Over complete line periods, orthogonality gives

$$\langle \sin\omega t \sin(n\omega t - \phi_n) \rangle = \tfrac{1}{2}\cos\phi_1\,\delta_{n1}, \tag{28}$$

where $\delta_{n1} = 1$ for $n = 1$ and zero otherwise. Therefore,

$$P_{\text{in}} = V_s I_1 \cos\phi_1, \qquad I_s^2 = I_{\text{DC}}^2 + \sum_{n=1}^{\infty} I_n^2, \tag{29}$$

$$\boxed{\text{PF} = \underbrace{\cos\phi_1}_{\text{DPF}}\ \underbrace{\frac{I_1}{I_s}}_{\text{DF}}\,.} \tag{30}$$

DPF measures fundamental phase alignment; DF measures the fraction of RMS current carried by the fundamental. Harmonic and switching currents increase RMS without contributing real power under a sinusoidal voltage.

C. Load-Sweep Conditions

For load fraction λ and rated power P_N,

$$P_{o,\lambda} \simeq \lambda P_N, \quad R_\lambda = \frac{V_{\text{set}}^2}{\lambda P_N}, \quad G_\lambda \simeq \frac{\lambda P_N}{\eta_\lambda V_s^2}, \tag{31}$$

$$I_{1,\lambda} = \frac{P_{\text{in},\lambda}}{V_s \cos\phi_{1,\lambda}}, \qquad I_{s,\lambda} = \frac{P_{\text{in},\lambda}}{V_s \text{PF}_\lambda}. \tag{32}$$

Set $\lambda = 0.25, 0.50, 0.75, 1.00$ while retaining the power stage and compensation. Measure settled active data over 780–800 ms with a 20 ns export grid [5].

TABLE III
LOAD-SWEEP MEASUREMENTS

Load / %	R / Ω	P_o / W	P_{in} / W	I_s / A	PF
25	577.60	250.09	253.32	1.1594	0.94993
50	288.80	500.19	507.51	2.2402	0.98500
75	192.53	750.33	763.14	3.3433	0.99244
100	144.40	1000.52	1020.25	4.4574	0.99516

TABLE IV
PF DECOMPOSITION AND CURRENT DISTORTION

Load / %	DPF	DF	THD_{40} / %	$I_{>40}$ / A
25	0.99485	0.95485	5.878	0.3382
50	0.99785	0.98712	4.388	0.3450
75	0.99857	0.99386	3.914	0.3463
100	0.99888	0.99628	3.714	0.3470

D. Why PF Deteriorates at Light Load

Using the harmonic-band residual defined in point 3,

$$\text{DF} = \left[1 + \text{THD}_{40}^2 + \frac{I_{\text{DC}}^2 + I_{>40}^2}{I_1^2}\right]^{-1/2}. \qquad (33)$$

The measured $I_{>40}$ remains 0.338–0.347 A RMS while I_1 falls from 4.4408 to 1.1071 A. Thus the nonfundamental-current fraction increases as load falls. The fundamental also leads the source: $\phi_1 = -5.816°$ at 25% versus $-2.710°$ at full load. Equations (30)–(33) account for PF changing from 0.99516 to 0.94993.

Figure 9 shows increasing relative ripple at light load. The PF decomposition separates displacement and distortion effects. Reference offsets, local DCM, and bridge/C_1 dynamics can contribute. Since C_1 is on the rectified side, a permanently connected AC-capacitor reactive-current formula is not used for this topology.

E. Waveforms and Interpretation

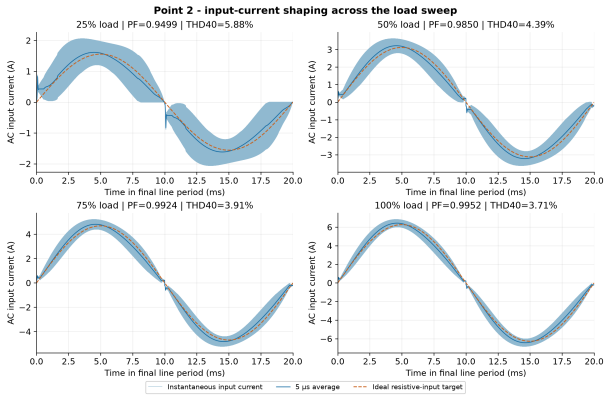


Fig. 10. Simulated load-current waveforms. The dashed resistive-input target is $(P_{\text{in}}/V_s^2)v_s$, rather than the internal multiplier reference.

The smaller useful fundamental at light load makes the nearly fixed residual current more prominent. These waveforms show the same load-dependent effect as the native SIMPLIS comparison.

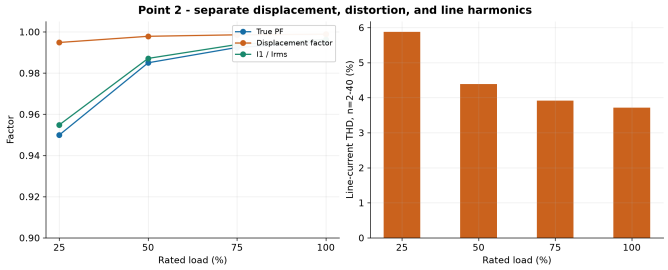


Fig. 11. Simulated displacement and distortion factors. PF falls from 0.99516 at full load to 0.94993 at 25% load.

Both DPF and DF decrease at light load. The larger DF reduction identifies the increasing nonfundamental-current fraction as the principal measured contribution in this model.

IV. Point 3: Harmonic Distortion

A. Harmonic Coefficients and Measurement Bandwidth

Define the complex Fourier coefficient over complete line periods:

$$c_n = \frac{1}{T_w}\int_{t_0}^{t_0+T_w} i_s(t)e^{-jn\omega t}\,\mathrm{d}t, \qquad I_n = \sqrt{2}|c_n| \quad (n \geq 1). \tag{34}$$

Here I_n is RMS, not peak amplitude. The reported distortion is

$$\mathrm{THD}_{40} = \frac{\sqrt{\sum_{n=2}^{40} I_n^2}}{I_1}, \qquad 40 f_{\mathrm{line}} = 2\,\mathrm{kHz}, \tag{35}$$

$$I_{>40} = \sqrt{I_s^2 - I_{\mathrm{DC}}^2 - \sum_{n=1}^{40} I_n^2}. \tag{36}$$

The residual contains represented current outside the first 40 coherent line harmonics, dominated here by switching ripple. PF uses the full RMS current. Only with negligible DC and THD over *all* nonfundamental content is

$$\mathrm{PF} = \frac{\mathrm{DPF}}{\sqrt{1+\mathrm{THD}_{\mathrm{all}}^2}}. \tag{37}$$

Substitution of THD_{40} into this identity would omit (36).

B. Symmetry and Zero-Crossing Distortion

For half-wave symmetry, $i_s(t+T/2) = -i_s(t)$. Splitting the integral into half cycles gives

$$c_n = \frac{1-(-1)^n}{T}\int_0^{T/2} i_s(t)e^{-jn\omega t}\,\mathrm{d}t. \tag{38}$$

Consequently, even harmonics vanish ideally; odd harmonics remain permitted.

For a simple crossing-distortion example, let $i_s = I_m \sin\theta$ for $\alpha \leq \theta \leq \pi - \alpha$ and zero elsewhere in the positive half cycle, extended with half-wave symmetry. Its sine coefficients are

$$b_n = \frac{2I_m}{\pi}\int_{\alpha}^{\pi-\alpha} \sin\theta \sin(n\theta)\,\mathrm{d}\theta. \tag{39}$$

Using $2\sin\theta\sin n\theta = \cos((n-1)\theta) - \cos((n+1)\theta)$,

$$b_1 = \frac{I_m}{\pi}(\pi - 2\alpha + \sin 2\alpha), \tag{40}$$

$$b_n = \frac{2I_m}{\pi}\left[\frac{\sin((n+1)\alpha)}{n+1} - \frac{\sin((n-1)\alpha)}{n-1}\right], \tag{41}$$

for $n = 3, 5, \ldots$, with $I_n = |b_n|/\sqrt{2}$. Crossing defects therefore create odd harmonics. This clipped-sine example explains the mechanism; it is not fitted to the UC3854 reference floor.

C. Inductance and Switching-Frequency Effects

From (16), under ideal CCM at fixed line phase,

$$\frac{\partial \ln \Delta i_L}{\partial \ln L} = -1, \qquad \frac{\partial \ln \Delta i_L}{\partial \ln f_s} = -1. \tag{42}$$

A zero-mean triangular switching ripple has

$$i_{\mathrm{rip,rms}}^2 = \frac{1}{\Delta i_L}\int_{-\Delta i_L/2}^{\Delta i_L/2} x^2\,\mathrm{d}x = \frac{\Delta i_L^2}{12}. \tag{43}$$

Line-cycle averaging gives

$$I_{\mathrm{HF}}^2 \simeq \frac{\langle v_r^2(1-v_r/V_o)^2\rangle}{12(Lf_s)^2}. \tag{44}$$

With $v_r = V_m|\sin\theta|$ and $k = V_m/V_o$,

$$I_{\mathrm{HF}}^2 \simeq \frac{V_m^2}{12(Lf_s)^2}\left(\frac{1}{2} - \frac{8k}{3\pi} + \frac{3k^2}{8}\right). \tag{45}$$

This follows from $\langle \sin^2\theta\rangle = 1/2$, $\langle|\sin\theta|^3\rangle = 4/(3\pi)$, and $\langle \sin^4\theta\rangle = 3/8$.

For local device/winding-drop corrections, define

$$V_{\mathrm{on}} = v_r - V_Q - r_L I, \qquad V_{\mathrm{off}} = v_r - v_o - V_D - r_L I. \tag{46}$$

Volt-second balance and the ON slope give

$$D = -\frac{V_{\mathrm{off}}}{V_{\mathrm{on}} - V_{\mathrm{off}}}, \qquad \Delta i_L \simeq \frac{V_{\mathrm{on}} D}{Lf_s}. \tag{47}$$

V_Q, V_D are local switch/diode drops; r_L is winding resistance. The ideal inverse-Lf_s ripple law does not imply monotonic low-order THD, which also depends on control, offsets, DCM, and crossing dynamics.

TABLE V
Full-Load Parameter Sweep

L / μH	f_s / kHz	PF	THD_{40} / %	Δi_L / A	Ideal / A
300	200	0.99516	3.714	0.8597	0.7815
150	200	0.98633	3.713	1.7200	1.5630
600	200	0.99740	3.732	0.4299	0.3908
300	100	0.98619	3.645	1.7206	1.5630
300	250	0.99625	3.742	0.6877	0.6252

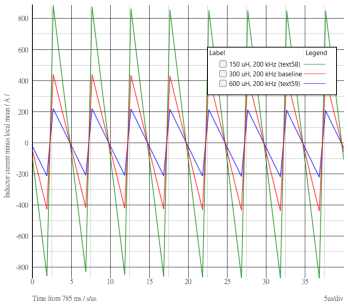


Fig. 12. SIMPLIS inductance-sweep switching ripple. Crest-ripple measurements are listed in Table V.

Doubling L halves the measured crest ripple, $0.8597 \to 0.4299$ A pp. Halving f_s doubles it, $0.8597 \to 1.7206$ A pp. In contrast, THD_{40} remains approximately 3.6–3.8% without a monotonic trend. True PF improves mainly as the represented switching RMS decreases, consistent with (45).

D. Dominant Harmonics

TABLE VI
DOMINANT BASELINE CURRENT HARMONICS

n	nf_{line} / Hz	I_n / A RMS	$100I_n/I_1$ / %
3	150	0.162503	3.6593
5	250	0.020621	0.4644
7	350	0.010891	0.2452
9	450	0.006756	0.1521
39	1950	0.005449	0.1227

At baseline, $I_3 = 0.162503\,\text{A}$ RMS (3.6593% of I_1); higher odd terms are smaller. Symmetry selects odd orders; their amplitudes require simulation or a complete waveform model.

E. Coherent Integration and Numerical Checks

SIMPLIS event samples are generally nonuniform. For linear interpolation of segment endpoints (x_0, y_0) and (x_1, y_1) over duration h,

$$\int_0^h y^2\,\mathrm{d}t = \frac{h}{3}(y_0^2 + y_0y_1 + y_1^2), \tag{48}$$

$$\int_0^h xy\,\mathrm{d}t = \frac{h}{6}(2x_0y_0 + x_0y_1 + x_1y_0 + 2x_1y_1). \tag{49}$$

Substitute $x = x_0 + u(x_1 - x_0)$, $y = y_0 + u(y_1 - y_0)$, and $\mathrm{d}t = h\,\mathrm{d}u$; integrate $u \in [0, 1]$.

For $i_s(t_k + \tau) = y_k + m_k\tau$, $0 \le \tau \le h_k$, set $z = jn\omega$. Direct integration yields

$$c_n = \frac{1}{T_w}\sum_k e^{-zt_k}(y_kA_k + m_kB_k), \tag{50}$$

$$A_k = \frac{1 - e^{-zh_k}}{z}, \qquad B_k = \frac{1 - e^{-zh_k}(1 + zh_k)}{z^2}. \tag{51}$$

These expressions exactly integrate the interpolated data. Evaluate small arguments stably. Tables use coherent integration; FFT requires uniform samples.

TABLE VII
SIMULATION-DATA VALIDATION

Check	Verified result
Settling, preceding five cycles	Maximum cycle-mean spans: V_o 0.000045%; P_{in} 0.000902%
25% load: 100 ns to 20 ns grid	ΔPF = 0.000093; ΔTHD_{40} = 0.00171 percentage points
Power-factor identity	Maximum $\lvert\text{PF} - \text{DPF}\cdot\text{DF}\rvert < 6.4\times10^{-15}$
Model run	43 top-level terminal mappings checked; ordinary F9 run completed

F. Parameter and Harmonic Comparisons

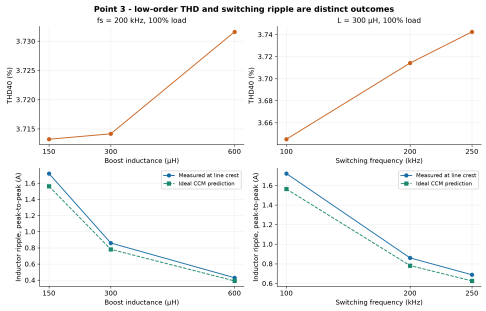


Fig. 13. Full-load parameter sweeps with fixed compensation; shared baseline: 300 μH, 200 kHz.

Ripple follows approximately $1/(Lf_s)$; nearly constant low-order THD reflects different distortion mechanisms.

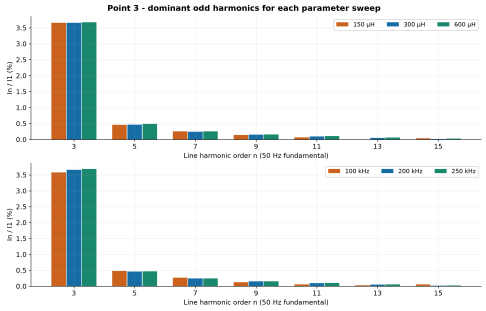


Fig. 14. Odd harmonics normalized to each fundamental. Numerical THD includes orders 2–40.

The third harmonic dominates. Changing L or f_s reduces switching ripple without removing reference-shape and crossing contributions.

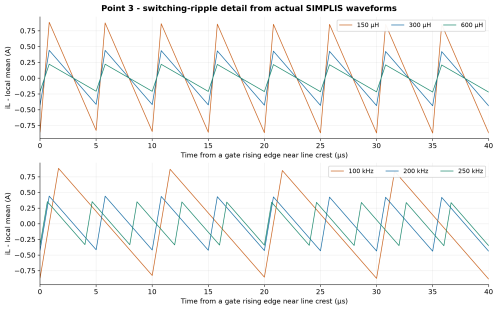


Fig. 15. Inductance/frequency ripple sweeps, aligned to gate rising edges with local means removed.

V. Point 4: With/Without-PFC Comparison

A. Passive Rectifier Architecture

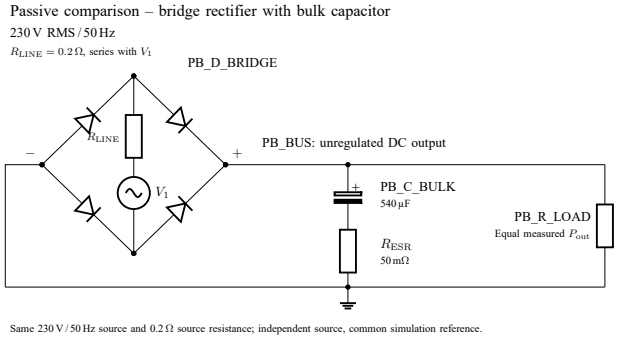


Fig. 16. Passive diode-bridge/bulk-capacitor comparator. No boost stage or voltage regulator is present.

The comparator uses the same 230 V RMS, 50 Hz source, $0.2\,\Omega$ source resistance, and $540\,\mu$F bulk capacitor with $50\,\mathrm{m}\Omega$ ESR. Its load is calibrated for equal measured output power.

To avoid confusion with the active boost-input node, define $u_r = |v_s|$ as the ideal rectified *source* envelope and v_b as the passive bus. A two-diode conducting path gives

$$j = \max\left(\frac{u_r - 2V_f - v_b}{R_s + 2r_f}, 0\right), \qquad i_{s,b} = \mathrm{sgn}(v_s)j, \quad (52)$$

$$C\dot{v}_b = j - \frac{v_b}{R_b}. \quad (53)$$

V_f, r_f model one bridge diode. The closed-form derivation neglects capacitor ESR; the switching simulation includes it. When the bridge blocks,

$$j = 0, \qquad v_b(t) = v_b(t_0)e^{-(t-t_0)/(R_bC)}. \quad (54)$$

Conduction begins when the source exceeds the capacitor voltage plus diode drops, creating short charging pulses instead of sinusoidal current.

B. Ideal Conduction Angles and PF

For $R_s = V_f = r_f = 0$, during the positive half cycle,

$$v_b = V_m \sin\theta, \qquad \alpha \le \theta \le \beta, \quad (55)$$

where α and β are turn-on and turn-off angles. Substitution into (53) gives

$$j = \omega C V_m \cos\theta + \frac{V_m}{R_b}\sin\theta. \quad (56)$$

At turn-off, $j(\beta) = 0$; during discharge the next turn-on occurs at $\pi + \alpha$. Therefore,

$$\beta = \pi - \tan^{-1}(\omega R_b C), \quad (57)$$

$$\sin\alpha = \sin\beta\, e^{-(\pi+\alpha-\beta)/(\omega R_bC)}. \quad (58)$$

Solve these together; do not assume an arbitrary pulse width. Extending the current with half-wave symmetry yields

$$I_{s,b}^2 = \frac{1}{\pi}\int_\alpha^\beta j^2(\theta)\,\mathrm{d}\theta, \quad (59)$$

$$P_{\mathrm{in},b} = \frac{V_m}{\pi}\int_\alpha^\beta j(\theta)\sin\theta\,\mathrm{d}\theta, \quad (60)$$

$$\mathrm{PF}_b = \frac{P_{\mathrm{in},b}}{(V_m/\sqrt{2})I_{s,b}}. \quad (61)$$

The harmonic coefficients follow directly from

$$a_n = \frac{1}{\pi}\int_0^{2\pi} i_{s,b}(\theta)\cos(n\theta)\,\mathrm{d}\theta, \qquad b_n = \frac{1}{\pi}\int_0^{2\pi} i_{s,b}(\theta)\sin(n\theta)\,\mathrm{d}\theta, \quad (62)$$

$$I_n = \frac{\sqrt{a_n^2 + b_n^2}}{\sqrt{2}}. \quad (63)$$

For finite source resistance and diode drops, use the simulated waveform in these power/harmonic integrals rather than forcing the ideal conduction angles onto the circuit.

C. DC-Bus Ripple from Energy Balance

For ideal unity-PF operation,

$$p_{\mathrm{in}} = V_m I_m \sin^2\theta = P_o(1 - \cos 2\omega t). \quad (64)$$

Multiplying the two averaged boost equations (6) by I and v_o and adding cancels the internal transfer term:

$$\frac{\mathrm{d}}{\mathrm{d}t}\left(\frac{LI^2}{2} + \frac{Cv_o^2}{2}\right) = v_r I - p_o(t), \qquad p_o(t) = \frac{v_o^2}{R}. \quad (65)$$

Neglecting inductor-energy variation and taking $p_o(t) \simeq P_o$,

$$\frac{\mathrm{d}}{\mathrm{d}t}\left(\frac{Cv_o^2}{2}\right) \simeq -P_o\cos 2\omega t, \quad (66)$$

$$v_o^2 \simeq V_o^2 - \frac{P_o}{\omega C}\sin 2\omega t. \quad (67)$$

Linearizing $v_o^2 \simeq V_o^2 + 2V_o\widetilde{v}_o$ gives

$$\widetilde{v}_o \simeq -\frac{P_o}{2\omega C V_o}\sin 2\omega t, \qquad \boxed{\Delta v_{o,pp} \simeq \frac{P_o}{\omega C V_o}}. \quad (68)$$

The corresponding low-frequency capacitor current is

$$i_{C,2\omega} \simeq -\frac{P_o}{V_o}\cos 2\omega t, \qquad I_{C,2\omega,\mathrm{rms}} \simeq \frac{P_o}{\sqrt{2}V_o}. \quad (69)$$

For the passive bus, approximately constant discharge current gives

$$\Delta v_{b,pp} \simeq \frac{I_o t_{\rm off}}{C} \simeq \frac{P_o}{2 f_{\rm line} C V_b}. \tag{70}$$

The last estimate assumes nearly a half-cycle discharge; finite conduction time reduces the actual ripple. Neither ideal formula includes ESR or a complete loss/current-distortion model.

D. Equal-Power Comparison

Use identical AC sources and matched measured output powers, not equal load resistances:

$$P_o = \frac{\langle v_o^2 \rangle}{R}, \qquad \epsilon_P = 100 \frac{P_{o,b} - P_{o,a}}{P_{o,a}} \quad (\%). \tag{71}$$

A practical passive-load iteration is

$$R_{b,\rm new} \simeq R_{b,\rm old} \frac{P_{b,\rm old}}{P_{a,\rm target}}, \tag{72}$$

followed by remeasurement. The passive bridge cannot regulate to 380 V from this source; equal output voltage would require another conversion stage.

TABLE VIII
MATCHED OUTPUT-POWER CONDITIONS

Load / %	R_b / Ω	$P_{o,a}$ / W	$P_{o,b}$ / W	ϵ_P / %
25	401.178	250.09	250.10	0.0046
50	192.890	500.19	500.17	-0.0054
75	123.821	750.33	750.01	-0.0423
100	89.478	1000.52	999.82	-0.0697

TABLE IX
MATCHED-POWER PF AND THD_{40}

Load / %	Active PF	Passive PF	Active THD / %	Passive THD / %
25	0.94993	0.42292	5.878	210.814
50	0.98500	0.48284	4.388	175.675
75	0.99244	0.52202	3.914	155.406
100	0.99516	0.55149	3.714	141.081

TABLE X
DC BUS AND RIPPLE AT MATCHED POWERS

Load / %	$V_{o,a}$ / V	V_b / V	$\Delta v_{o,a}$ / V	Δv_b / V
25	380.07	316.73	4.10	12.99
50	380.06	310.52	8.15	25.33
75	380.06	304.53	12.16	37.27
100	380.06	298.73	16.16	48.90

TABLE XI
PEAK-TO-PEAK RIPPLE: MEASURED AND ESTIMATED

Load / %	Active meas. / V	Active ideal / V	Passive meas. / V	Passive est. / V
25	4.10	3.88	12.99	14.62
50	8.15	7.76	25.33	29.83
75	12.16	11.64	37.27	45.61
100	16.16	15.52	48.90	61.98

Full-load active/passive results: PF $0.99516/0.55149$; $\mathrm{THD}_{40} = 3.714/141.081\%$; bus ripple $16.16/48.90$ V pp; peak source current $6.83/25.93$ A. Active regulation requires an inductor, switch, diode, sensing, and control, with switching/drive/EMI and thermal design.

E. Engineering Reference: TI PMP12074

TABLE XII
ENGINEERING REFERENCE: 230 V/50 HZ TESTS

Sim. P_o / W	Sim. PF	TI P_o / W	TI PF	TI η / %
250.09	0.94993	228.060	0.993	95.26
500.19	0.98500	475.250	0.997	96.85
750.33	0.99244	760.600	0.999	97.11
1000.52	0.99516	1007.795	0.999	97.37

TABLE XIII
FULL-LOAD ACTIVE/PASSIVE TRADEOFFS

Quantity	Active PFC	Passive bridge
RMS / peak source current	4.457 / 6.83 A	8.079 / 25.93 A
DC bus / ripple	380.06 V / 16.16 V pp	298.73 V / 48.90 V pp
Hardware	Bridge, L, switch, diode, controller	Bridge, bulk capacitor
Tradeoff	Regulation and improved current quality; switching, drive, losses and EMI	Fewer components; pulsed current, unregulated bus

TABLE XIV
ENGINEERING-REFERENCE COMPARISON SCOPE

Item	SIMPLIS study	TI PMP12074
Topology / rating	Bridge + CCM boost; 230 V to 380 V, 1 kW, 200 kHz	Same broad topology/rating
Controller	UC3854; explicit line multiplier and feedforward	UCC28180
Devices / sensing	IRF450, MUR860; 0.1 Ω shunt	Different devices; 0.04 Ω shunt
Filtering / evidence	330 nF after bridge; switching simulation	EMI network; hardware measurements

PMP12074 is a CCM boost reference at the same broad rated point, but uses UCC28180 instead of UC3854 [2]. Its published 230 V/50 Hz test points [3] are near, rather than exactly at, this study's quarter-load powers. Controller, devices, filtering, and measurement bandwidth differ. The simulation does not validate TI efficiency or regulatory compliance.

F. Waveforms and Interpretation

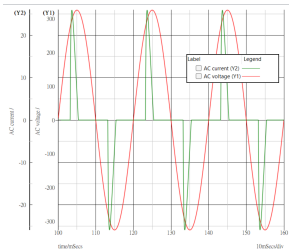


Fig. 17. Simulated passive voltage and current (100–160 ms). Charging current occurs near source-voltage crests.

Between charging intervals, the capacitor supplies the load and its voltage falls.

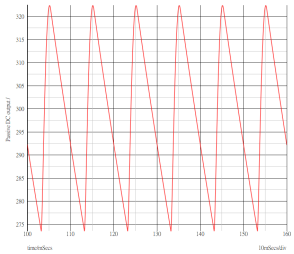


Fig. 18. Simulated passive bus (100–160 ms). The waveform illustrates capacitor discharge; steady-state measurements use 200–300 ms.

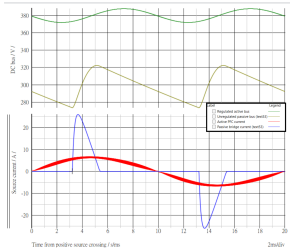


Fig. 19. Active/passive comparison: regulated 380 V bus and distributed current versus capacitor discharge and crest-concentrated charging pulses.

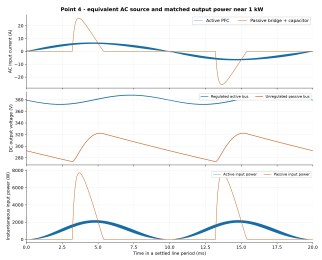


Fig. 20. Simulated matched full-load currents, buses, and instantaneous input powers. Curves are phase-aligned for display.

Passive charging is pulsed; the capacitor supplies the intervening load. Active PFC spreads input power over the cycle, reducing peaks and improving PF.

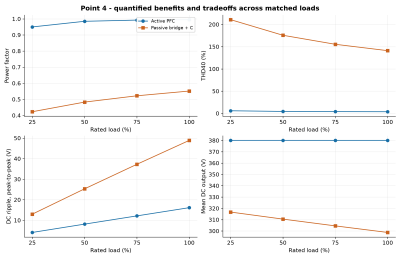


Fig. 21. Simulated matched-power PF, THD, mean bus, and ripple across the load sweep.

Active PFC maintains mean bus voltage and lower THD. Passive output is unregulated, with stronger load-dependent ripple.

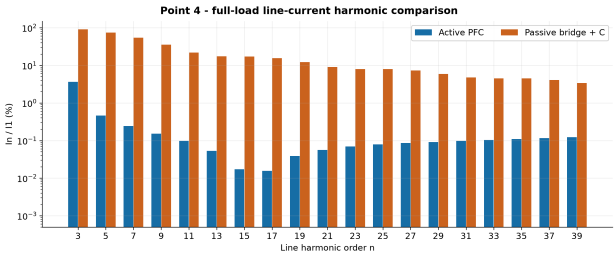


Fig. 22. Simulated full-load harmonic comparison. It displays odd harmonics; the numerical THD includes all orders 2–40.

VI. Point 5: Voltage-Loop Control

A. Circuit, Operating Point, and Notation

v: instantaneous bus; V: equilibrium bus; $\bar{v}$: slow mean; $I = \langle i_L \rangle_s$; d: ON duty; u: voltage-amplifier output. Hats denote perturbations. $\omega = 2\pi \cdot 50\,\mathrm{rad/s}$ and $\Omega_r = 2\omega$. Ripple amplitudes are peak values unless marked RMS or peak-to-peak.

$q = \eta p$ is effective power delivered to the bus. Define $K_u = \partial p/\partial u$ and $K = \eta K_u = \partial q/\partial u$ in W/V. Assume local CCM, small ripple, unsaturated operation, and a faster current loop. The simulation retains circuit nonidealities.

TABLE XV
Point-5 Circuit and Test Boundary

Quantity	Value
Source / PWM	230 V RMS, 50 Hz; 200 kHz; $R_s = 0.2\,\Omega$
Energy storage	$L_2 = 300\,\mu$H; $C_2 = 540\,\mu$F; $C_1 = 330\,$nF
Setpoint / rating	$V = 380\,$V; $P = 1\,$kW
Sensing	$R_3 = 0.1\,\Omega$; $I^* \simeq 40000 I_{\mathrm{mult}}$
Divider	$R_t = R_7 = 496.6667\,$kΩ; $R_b = R_8 = 10\,$kΩ; $V_{\mathrm{ref}} = 7.5\,$V
Compensation	$R_z = 5.6\,$kΩ; $C_z = C_8 = 4.7\,\mu$F; $R_f = R_{14} = 100\,$MΩ
Load switch	$288.8\,\Omega$ base; add $288.8\,\Omega + 1\,$mΩ switch
Initialization	Precharged bus and seeded states; no cold-start test.

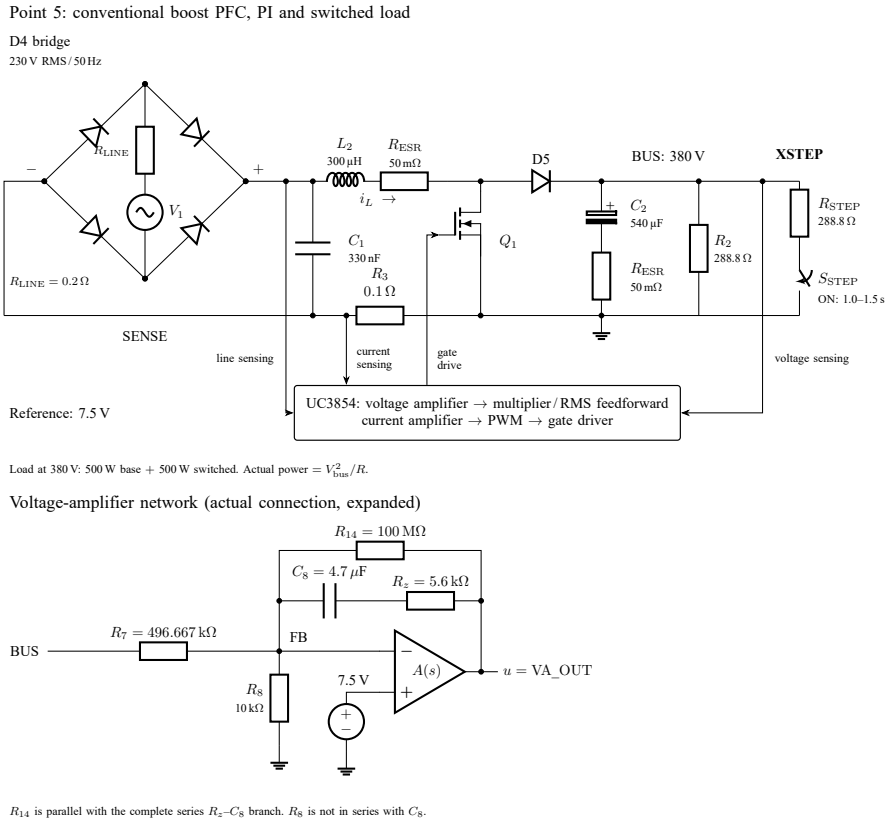


Fig. 23. Standard bridge–CCM boost PFC, UC3854 controller, actual compensation network, and switched resistive load.

PMP12074 remains a topology/rating reference. The simulated controller is a UC3854 average-current macromodel, rather than an exact UCC28180 implementation [1], [2], [7].

B. Switching States, Averaging, and the Energy Plant

$$Q\ \mathrm{ON}: \quad L\dot{i}_L = v_r, \quad C\dot{v} = -v/R, \tag{5.1}$$

$$Q\ \mathrm{OFF}: \quad L\dot{i}_L = v_r - v, \quad C\dot{v} = i_L - v/R. \tag{5.2}$$

$$L\dot{I} = v_r - (1-d)v, \tag{5.3}$$

$$C\dot{v} = (1-d)I - v/R. \tag{5.4}$$

KVL/KCL give the ON/OFF states. Weighting by d and $1-d$ removes switching ripple while retaining line-frequency variation. Linearize about a frozen CCM point with $\alpha = 1-D$ and independent load-current disturbance $\widehat{i}_\ell$:

$$Ls\widehat{I} = \widehat{v}_r - \alpha\widehat{v} + V\widehat{d} \tag{5.5}$$

$$(Cs + 1/R)\widehat{v} = \alpha\widehat{I} - I_0\widehat{d} - \widehat{i}_\ell \tag{5.6}$$

$$\mathcal{D}(s) = LCs^2 + (L/R)s + \alpha^2 \tag{5.7}$$

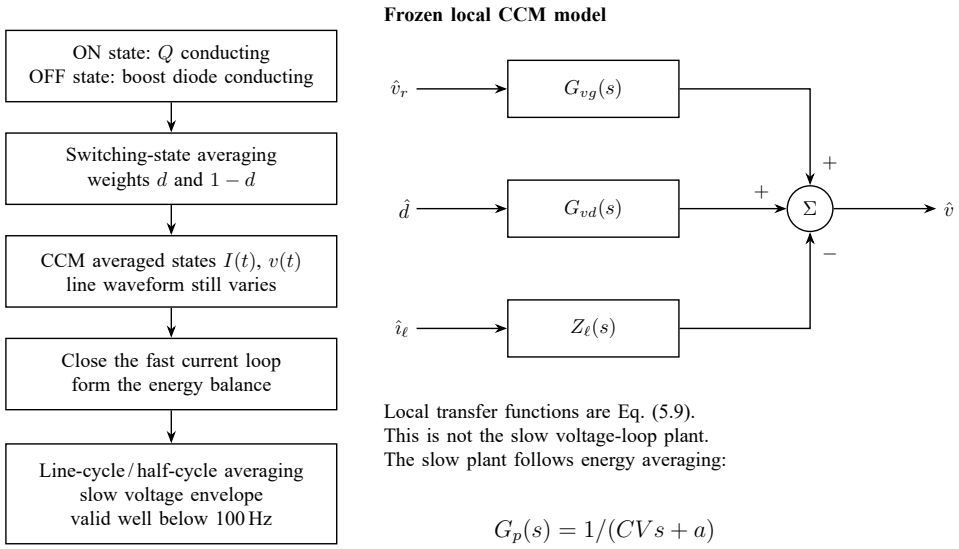
$$\widehat{v} = G_{vg}\widehat{v}_r + G_{vd}\widehat{d} - Z_\ell\widehat{i}_\ell \tag{5.8}$$

$$G_{vg} = \frac{\alpha}{\mathcal{D}}, \qquad G_{vd} = \frac{\alpha V - LI_0 s}{\mathcal{D}}, \qquad Z_\ell = \frac{Ls}{\mathcal{D}} \tag{5.9}$$

$$\omega_{z,\mathrm{RHP}} = \frac{\alpha V}{LI_0} = \frac{R\alpha^2}{L} \quad \text{(local equilibrium)} \tag{5.10}$$

Eliminating $\widehat{I}$ gives the three disturbance paths. The RHP zero belongs to this frozen duty plant; it is not directly the current-regulated PFC voltage-loop plant.

Two averaging levels and the frozen CCM perturbation model



Local transfer functions are Eq. (5.9).
This is not the slow voltage-loop plant.
The slow plant follows energy averaging:

$$G_p(s) = 1/(CVs + a)$$

A frozen DC linearization cannot replace the periodically varying PFC operating trajectory.

Fig. 24. Two averaging levels and the frozen CCM disturbance paths. Local plant paths are defined in (5.9).

$$\frac{d}{dt}\left(\frac{LI^2}{2} + \frac{Cv^2}{2}\right) = v_r I - \frac{v^2}{R} \tag{5.11}$$

$$\frac{d}{dt}\left(\frac{C\bar{v}^2}{2}\right) \simeq q - P_\ell(\bar{v}) \qquad (f \ll 2f_{\mathrm{line}}) \tag{5.12}$$

$$\overline{W}_L = \frac{Lp^2}{2V_s^2}, \quad \tau_L = \frac{Lp_0}{V_s^2} \tag{5.13}$$

$$(CVs + a)\widehat{v} = (\eta - \tau_L s)\widehat{p} - \widehat{p}_\ell \tag{5.14}$$

$$\tau_L \simeq 5.67\ \mu\mathrm{s} \quad (p_0 = 1000\ \mathrm{W},\ \eta = 1)$$

$$b\dot{\widehat{v}} + a\widehat{v} = \widehat{q} - \widehat{p}_\ell, \qquad b = CV \tag{5.15}$$

$$G_p(s) = \frac{\widehat{v}}{\widehat{q}} = \frac{1}{bs + a}, \quad Z_P(s) = \frac{1}{bs + a} \tag{5.16}$$

$$a = \frac{2V}{R} = \frac{2P}{V}, \quad f_p = \frac{a}{2\pi b} = \frac{1}{\pi RC} \quad \text{(resistor)} \tag{5.17}$$

Multiply the averaged inductor equation by I and the capacitor equation by v: the internal transfer terms cancel. A subsequent line average gives the slow energy plant. At 1 kW, $\overline{W}_L \simeq 2.84\,\mathrm{mJ}$ versus $W_C = 38.99\,\mathrm{J}$; $b = CV = 0.2052\,\mathrm{J/V}$, $a = 2P/V = 5.26316\,\mathrm{W/V}$, and $f_p = 4.082\,\mathrm{Hz}$.

For $P_\ell = gv^2$, $\delta P_\ell = (2P/V)\widehat{v} + V^2\widehat{g}$ and $\widehat{p}_\ell = V^2\widehat{g}$. The factor two is required for a resistor. A constant-power load instead gives $a = 0$; a constant-current load gives $a = I_o$.

C. Physical Compensation and PI Gain Mapping

$$H = \frac{R_b}{R_t + R_b} = \frac{7.5}{380} = 0.0197368 \tag{5.18}$$

$$R_{\mathrm{th}} = R_t \parallel R_b = HR_t = 9.80263\ \mathrm{k\Omega} \tag{5.19}$$

$$\begin{aligned} Z_f(s) &= R_f \parallel \left(R_z + \frac{1}{sC_z}\right) \\ &= \frac{R_f(1 + sR_zC_z)}{1 + s(R_f + R_z)C_z}. \end{aligned} \tag{5.20}$$

At fixed reference and high amplifier gain, $\widehat{v}_{\mathrm{FB}} \simeq 0$. KCL gives $\widehat{v}/R_t + \widehat{u}/Z_f = 0$, hence $\widehat{u} = -(Z_f/R_t)\widehat{v}$.

$$C_v(s) = \frac{Z_f(s)}{R_t} \simeq \frac{R_z}{R_t} + \frac{1}{sR_tC_z} \tag{5.21}$$

$$K_{pv} = 0.0112752\ \mathrm{V/V}, \quad K_{iv} = 0.428388\ \mathrm{V/(V\,s)} \tag{5.22}$$

$$f_z = \frac{1}{2\pi R_zC_z} = 6.0469\ \mathrm{Hz}, \quad f_{\mathrm{leak}} \simeq 0.0003386\ \mathrm{Hz} \tag{5.23}$$

$$\frac{\widehat{u}}{\widehat{v}} = -\frac{Z_f}{R_t}\,\frac{A(s)}{A(s) + 1 + Z_f/R_{\mathrm{th}}} \tag{5.24}$$

$$\begin{aligned} C_P(s) &= KC_v(s) = K_p + K_i/s, \\ K_p &= KR_z/R_t, \qquad K_i = K/(R_tC_z). \end{aligned} \tag{5.25}$$

K_{pv} and K_{iv} are bus-error gains. K_p and K_i are power-domain gains in W/V and W/(V·s). Do not multiply C_v by H again. R_{14} makes the integrator leaky; (5.24) includes finite amplifier gain $A(s)$.

$$\begin{aligned} J &= \mathcal{F}_m(u, I_{\mathrm{AC}}, V_{\mathrm{FF}}), \\ I^* &= A_sJ, \qquad A_s = R_6/R_3 = 40000. \end{aligned} \tag{5.25a}$$

$$K_u = A_s \left\langle v_r \frac{\partial \mathcal{F}_m}{\partial u} \right\rangle_{\mathrm{line}} \tag{5.25b}$$

$$\begin{aligned} K_u &\simeq \frac{A_sk_m}{R_{\mathrm{AC}}k_{\mathrm{FF}}^2}, \\ I_{\mathrm{AC}} &= v_r/R_{\mathrm{AC}}, \quad V_{\mathrm{FF}} = k_{\mathrm{FF}}V_s. \end{aligned} \tag{5.25c}$$

J is the positive multiplier-current magnitude. Equation (5.25c) additionally assumes $J = k_mI_{\mathrm{AC}}(u - U_{\mathrm{off}})/V_{\mathrm{FF}}^2$. Actual bias, feedforward, current tracking, and limits require the model law in (5.25b).

Scale $R_z(\lambda) = \lambda R_{z0}$ and $C_z(\lambda) = C_{z0}/\lambda$. Then $C_v(\lambda) \simeq \lambda C_{v0}$ while $f_z = 6.0469\,\mathrm{Hz}$ remains fixed. Only the outer compensation changes.

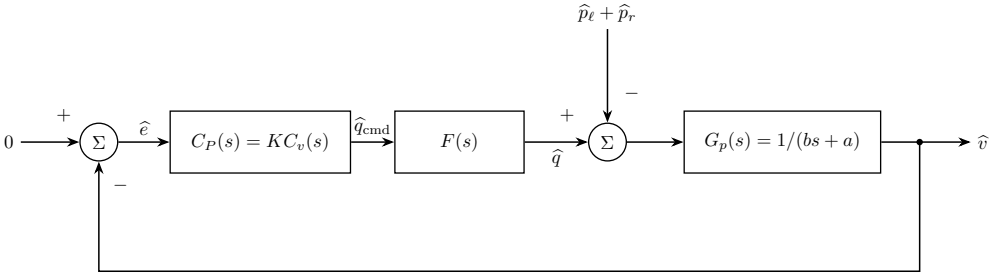
TABLE XVI
PHYSICAL PI SWEEP

Case / λ	R_z / kΩ	C_z / μF	K_{pv}	K_{iv} / s^{-1}
Slow / 0.35	1.960	13.428571	0.003946	0.149936
Baseline / 1	5.600	4.700000	0.011275	0.428388
Medium / 2.3	12.880	2.043478	0.025933	0.985292
Fast / 5	28.000	0.940000	0.056376	2.141939

D. Closed Loop, Crossover, and Stability

Outer-loop small-signal model in physical bus-voltage units

Fixed setpoint perturbation is zero; load and 100 Hz ripple are separate disturbances.



Unity feedback: physical bus voltage

$$T = C_PFG_p, \qquad S = \frac{1}{1+T}, \qquad T_c = \frac{T}{1+T}$$

$\widehat{p}_r \simeq P\cos(2\omega t)$ is restored only in the first-harmonic ripple calculation.

Equivalent sensed representation: $C_e = Z_f/R_{\mathrm{th}}$ with feedback H; $HC_e = C_v$. Do not count H twice.

Fig. 25. Physical-voltage feedback with $C_P = KC_v$, $G_p = 1/(bs + a)$, and residual current-loop dynamics $F(0) = 1$.

$$T = C_P F G_p, \qquad S = \frac{1}{1+T}, \qquad T_c = \frac{T}{1+T} \tag{5.26}$$

$$\frac{\widehat{v}}{\widehat{p}_\ell} = -G_p S, \qquad \frac{\widehat{q}}{\widehat{p}_\ell} = T_c, \qquad S + T_c = 1. \tag{5.27}$$

$$|T(j\Omega_c)| = 1, \qquad \mathrm{PM} = 180^\circ + \arg T(j\Omega_c) \tag{5.28}$$

$$b s^2 + (a + K_p) s + K_i = 0 \tag{5.29}$$

$$\omega_n = \sqrt{K_i / b}, \qquad \zeta = \frac{a + K_p}{2\sqrt{b K_i}} \tag{5.30}$$

$$K_i > 0, \quad a + K_p > 0 \quad \Rightarrow \quad \text{stable reduced model} \tag{5.31}$$

$$K_p = \frac{\sqrt{a^2 + (b\Omega_c)^2}}{|F(j\Omega_c)| \sqrt{1 + (\Omega_z/\Omega_c)^2}}, \qquad K_i = \Omega_z K_p. \tag{5.32}$$

$$\arg T = -\tan^{-1}\frac{\Omega_z}{\Omega_c} - \tan^{-1}\frac{b\Omega_c}{a} + \arg F \tag{5.33}$$

With $F \simeq 1$, $1 + C_P G_p = 0$ gives (5.29). Positive coefficients establish reduced-model stability only. Equation (5.32) enforces unity return ratio for $\Omega_z = K_i/K_p$. Crossover, natural frequency, and closed-loop $-3\,\mathrm{dB}$ bandwidth are distinct.

Inject a series perturbation between BUS and the top of R_7: $T(j\Omega) = -\widehat{V}_{\mathrm{bus}}/\widehat{V}_{\mathrm{sense_top}}$. Use $0.5\,\mathrm{V}$ peak per tone at $\{2, 4, 6, 8, 10, 12, 16, 20, 30, 40\}$ Hz; analyze 1.1–1.6 s and compare 0.6–1.1 s. Interpolate crossover and phase margin in log frequency. This measures the low-frequency diagonal response about a periodic point.

TABLE XVII
MEASURED VOLTAGE-LOOP RETURN RATIO

Case	f_c / Hz	PM / °	Window change / %
Slow	2.167	81.51	0.1415
Baseline	5.335	78.35	0.0103
Medium	10.830	80.51	0.0055
Fast	21.832	83.19	0.0104

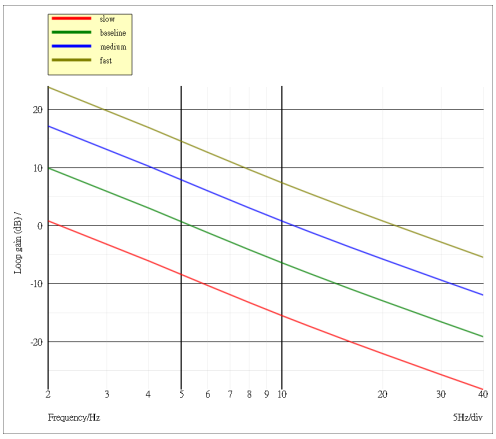


Fig. 26. Return-ratio gain for all PI settings. The unity-gain crossover moves from 2.167 to 21.832 Hz.

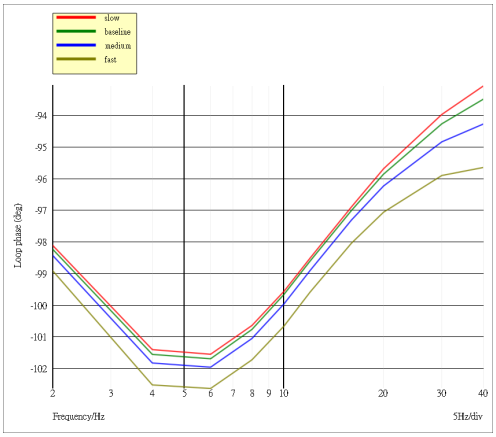


Fig. 27. Return-ratio phase; the measured crossover phase margins exceed 78°.

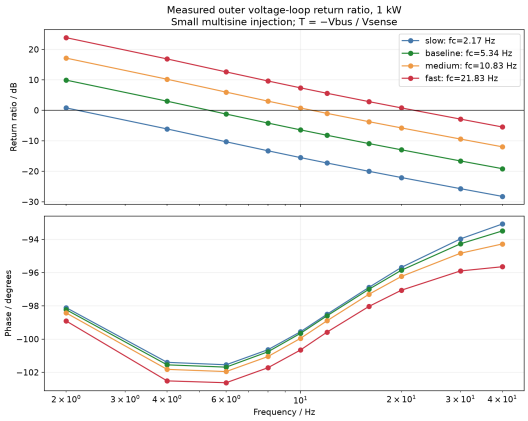


Fig. 28. Additional evidence: postprocessed SIMPLIS data. The same measured gain, phase, and crossover comparison.

E. Twice-Line Ripple and Third-Harmonic Generation

$$p_{\text{in}}(t) = P_{\text{in}}(1 - \cos 2\theta), \qquad \theta = \omega t \tag{5.34}$$

$$b\dot{\tilde{v}} + a\tilde{v} \simeq -P\cos\Omega_r t \tag{5.35}$$

$$V_{2,\text{pk}}^{\text{open}} = \frac{P}{\sqrt{a^2 + (b\Omega_r)^2}} \simeq \frac{P}{\Omega_r C V} \tag{5.36}$$

$$\underline{V}_2 \simeq -P\,G_p(j\Omega_r)S(j\Omega_r) \tag{5.37}$$

$$\widehat{Q}_2 \simeq P\,T_c(j\Omega_r), \quad m = \frac{|\widehat{Q}_2|}{P} \simeq |T_c(j\Omega_r)| \tag{5.38}$$

$$V_{2,\text{pk}} \simeq \frac{P|S(j\Omega_r)|}{\sqrt{a^2 + (b\Omega_r)^2}} \tag{5.39}$$

For $v_s = V_m \sin\theta$ and $i_s = GV_m \sin\theta$, input power contains a 100 Hz term. Equation (5.36) gives $V_{2,\text{pk}} \simeq 7.76\,\text{V}$ and $V_{2,\text{pp}} \simeq 15.51\,\text{V}$. Feedback reacts to this natural ripple. Equations (5.37)–(5.39) are first-harmonic estimates; sideband coupling is neglected, and the slow plant is not exact at 100 Hz.

Set $G(t) = G_0[1 + m\cos(2\theta + \psi)]$. Substitute $i_s = G(t)V_m \sin\theta$ and use $\sin A \cos B = [\sin(A+B) + \sin(A-B)]/2$:

$$\begin{aligned} i_s = I_m\big[&\sin\theta + \tfrac{m}{2}\sin(3\theta + \psi) \\ &- \tfrac{m}{2}\sin(\theta + \psi)\big]. \end{aligned} \tag{5.40}$$

$$\begin{aligned} I_{1,\text{pk}} &= I_m\sqrt{1 - m\cos\psi + m^2/4}, \\ I_{3,\text{pk}} &= I_m m/2. \end{aligned} \tag{5.41}$$

$$\text{THD}_3 = \frac{m/2}{\sqrt{1 - m\cos\psi + m^2/4}} \simeq \frac{m}{2} \quad (m \ll 1). \tag{5.42}$$

$$\text{DPF} = \frac{1 - (m/2)\cos\psi}{\sqrt{1 - m\cos\psi + m^2/4}} \tag{5.43}$$

$$\text{PF} = \frac{1 - (m/2)\cos\psi}{\sqrt{1 - m\cos\psi + m^2/2}} \tag{5.44}$$

$$|T_c(j\Omega_r)| \lesssim 2\epsilon \quad (\text{THD}_3 \le \epsilon,\ m \ll 1). \tag{5.45}$$

Second-harmonic amplitude modulation mixes into the first and third current harmonics. $\text{THD}_3 \simeq m/2$ applies only for small m and this isolated mechanism. Source-capacitor current, crossing error, and switching ripple remain; harmonic contributions add as phasors.

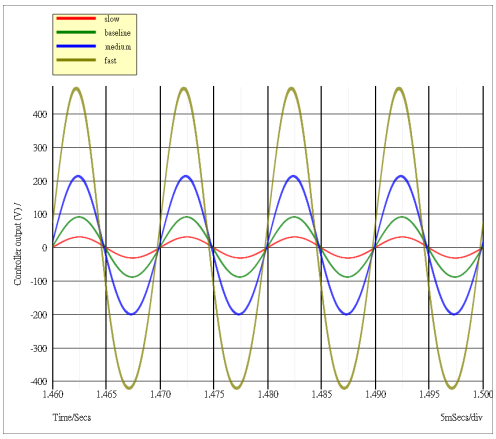


Fig. 29. 100 Hz voltage-amplifier output ripple. Its peak rises from 0.0314 V (slow) to 0.4472 V (fast). The multiplier offset and power slope are needed to infer m.

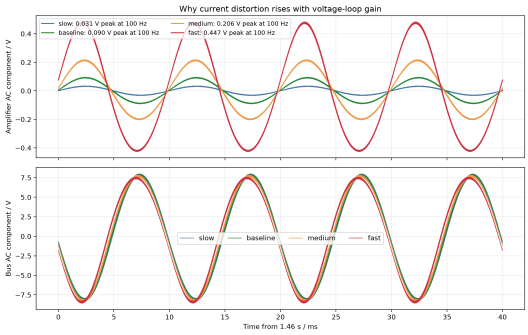


Fig. 30. Additional evidence: postprocessed SIMPLIS data. VA_OUT ripple increases with PI gain. Dividing ripple by raw VA_OUT does not give m.

F. Load-Step Equations and Nonlinear Energy Limits

$$\widehat{v}(s) = -\frac{\Delta P}{bs^2 + (a + K_p)s + K_i} \tag{5.46}$$

$$\dot{\hat{v}}(0^+) = -\frac{\Delta P}{CV}, \qquad \widehat{v}(\infty) = 0 \quad \text{(ideal stable PI)} \tag{5.47}$$

$$\begin{aligned} \widehat{v}(t) &= -\frac{\Delta P}{b\omega_d} e^{-\zeta\omega_n t}\sin\omega_d t, \\ \omega_d &= \omega_n\sqrt{1 - \zeta^2}. \end{aligned} \tag{5.48}$$

$$\begin{aligned} &\frac{C}{2}\big[v^2(t) - v^2(t_0)\big] \\ &= \int_{t_0}^{t} \big[p_{\text{bus}}(\tau) - P_\ell(\tau)\big]\,d\tau. \end{aligned} \tag{5.49}$$

$$C \ge \frac{2\Delta E_{\text{deficit}}}{V_0^2 - V_{\min}^2}, \quad V_{\min} < V_0 \tag{5.50}$$

$$v(t) \simeq \sqrt{V_0^2 - 2\Delta P(t - t_0)/C} \quad \text{(fixed deficit)}. \tag{5.51}$$

Apply $\widehat{p}_\ell = \Delta P/s$ to $-G_pS$. Equation (5.48) assumes $0 < \zeta < 1$; for distinct real poles use $-\Delta P[e^{r_1 t} - e^{r_2 t}]/[b(r_1 - r_2)]$. At $\Delta P = 500\,\mathrm{W}$, the initial envelope slope is $-2.44\,\mathrm{V/ms}$. Large steps require (5.49), including actual resistive power and line phase.

Saturation boundary: $\dot{x} = K_{iv}e$, $u_{\mathrm{cmd}} = K_{pv}e + x$, $u = \mathrm{sat}(u_{\mathrm{cmd}})$. Continued integration during limiting causes windup. Optional back-calculation adds $K_{aw}(u - u_{\mathrm{cmd}})$, with K_{aw} in s^{-1}; this addition is not claimed for the UC3854 macromodel.

G. Simulation Conditions and Measurement Definitions

TABLE XVIII
Point-5 Simulation Protocol

Test	Definition
PI sweep	$\lambda = 0.35, 1, 2.3, 5$; fixed stage, current loop, and f_z.
Large step	$500 \to 1000 \to 500\,\mathrm{W}$ rated resistive load at 1.0 / 1.5 s; stop 2 s.
Steady current	1.46–1.50 s: two 50 Hz cycles; minimum 20 ns export grid.
Transient	0.8–2.0 s; raw native extrema; compact traces at 1 μs.
Small step	$500 \to 550 \to 500\,\mathrm{W}$ at 0.2 / 0.5 s; settled 500 W initial state.
Line phase	Baseline large steps at 0° and 90°; no exhaustive worst-case claim.

$$c_n = \frac{1}{T_w}\int_{t_w}^{t_w+T_w} i_s(t)e^{-jn\omega t}\,dt, \quad I_n = \sqrt{2}|c_n|. \tag{5.52}$$

$$\mathrm{THD}_{2:40} = \frac{\sqrt{\sum_{n=2}^{40} I_n^2}}{I_1}, \quad \mathrm{PF} = \frac{\langle v_s i_s\rangle}{V_{s,\mathrm{rms}} I_{s,\mathrm{rms}}}. \tag{5.53}$$

$$\widetilde{v}_{\mathrm{env}}(t) = \frac{1}{10\,\mathrm{ms}}\int_{t-5\,\mathrm{ms}}^{t+5\,\mathrm{ms}} v(\tau)\,d\tau. \tag{5.54}$$

$$t_{\mathrm{rec}} = \inf\{\tau \ge 0 : |\widetilde{v}_{\mathrm{env}}(t) - 380| \le 3.8\,\mathrm{V} \quad \forall t \in [t_{\mathrm{step}} + \tau, t_{\mathrm{end}}]\}. \tag{5.55}$$

t_{end} is the next step or record end. Integrate Fourier coefficients on actual time coordinates using piecewise-linear interpolation. RMS includes represented switching content, so true PF need not equal $\mathrm{DPF}/\sqrt{1 + \mathrm{THD}_{2:40}^2}$. Recovery uses a centered, noncausal 10 ms analysis mean; report raw extrema separately. Measure steady THD only in settled windows.

H. Current Quality, Regulation, and Load Recovery

TABLE XIX
Settled Full-Load Results

Case	THD / %	PF	V_{DC} / V	Δv_{pp} / V
Slow	2.788	0.99604	380.094	16.174
Baseline	3.714	0.99516	380.057	16.159
Medium	6.179	0.99221	380.057	16.153
Fast	11.716	0.98153	380.057	16.131

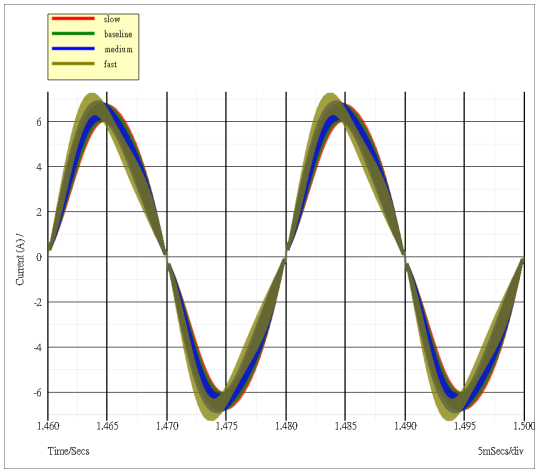


Fig. 31. Settled source current at 1 kW, 1.46–1.50 s. Faster PI changes the line-current shape.

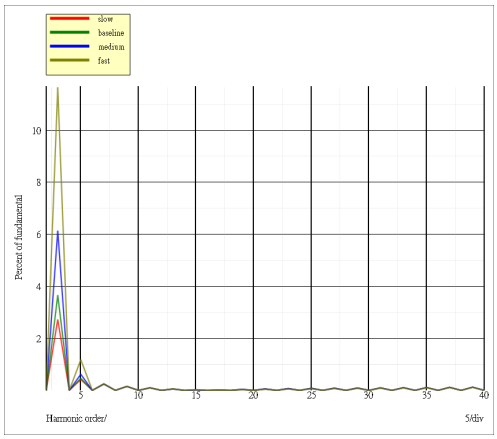


Fig. 32. Current harmonics for the four PI settings; faster voltage feedback raises low-order distortion.

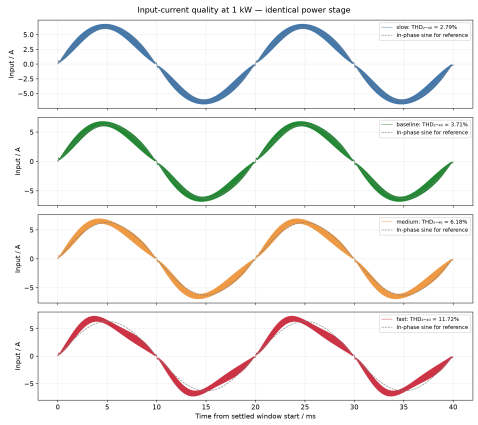


Fig. 33. Additional evidence: postprocessed SIMPLIS data. Individual case waveforms at the same AC input and power-stage conditions.

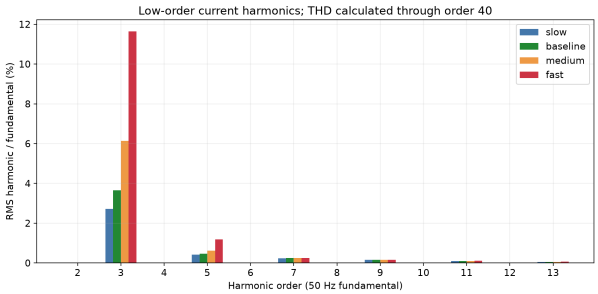


Fig. 34. Additional evidence: postprocessed SIMPLIS data. Measured current harmonics and increasing low-order distortion.

TABLE XX
LARGE-STEP BUS EXCURSION AND RECOVERY

Case	$v_{\min}$ / V	$v_{\max}$ / V	$t_{\uparrow}$ / ms	$t_{\downarrow}$ / ms
Slow	321.34	445.58	239.0	327.8
Baseline	340.45	420.17	120.0	110.2
Medium	352.33	405.20	75.8	73.1
Fast	360.21	395.83	48.0	48.2

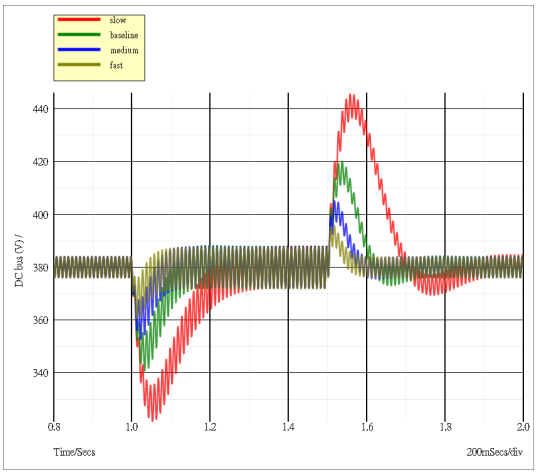


Fig. 35. Raw bus voltage for $500 \leftrightarrow 1000\,\mathrm{W}$ rated resistive-load steps. Faster PI reduces sag and overshoot.

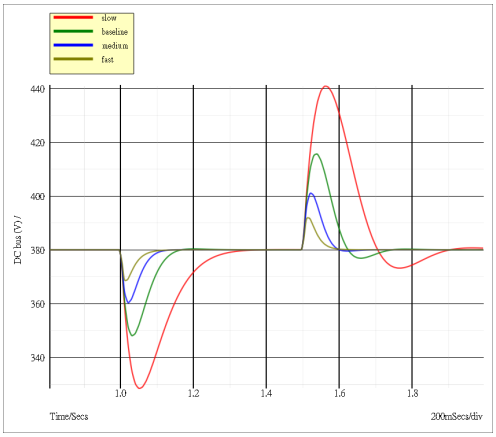


Fig. 36. 10 ms bus envelopes and the $380 \pm 3.8\,\mathrm{V}$ recovery band.

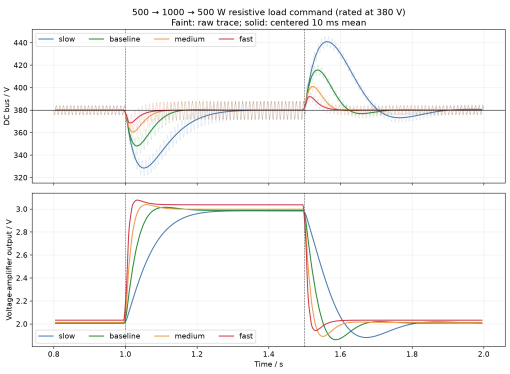


Fig. 37. Additional evidence: postprocessed SIMPLIS data. Raw bus and envelopes for the same load-step cases.

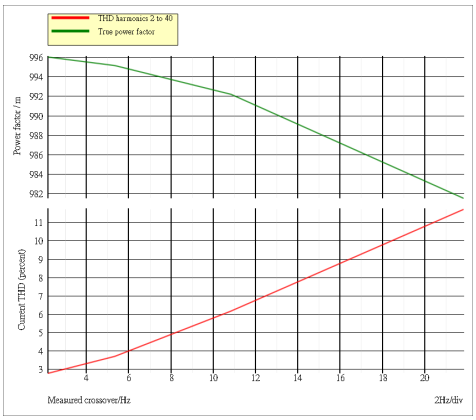


Fig. 38. Measured crossover versus current quality. The 5% THD comparison target is a comparison choice, not a certification limit.

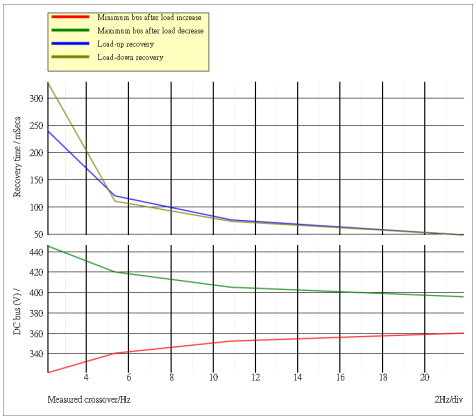


Fig. 39. Measured crossover versus regulation and recovery. Faster response trades current quality for smaller bus excursions.

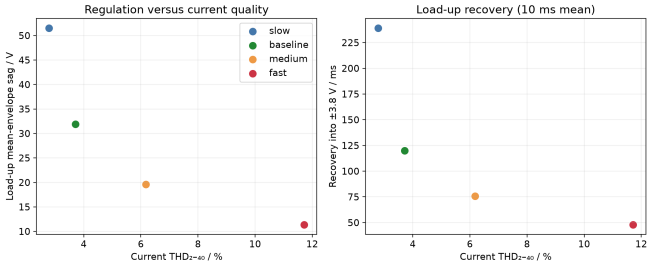


Fig. 40. Additional evidence: postprocessed SIMPLIS data. THD, measured crossover, excursion, and recovery comparison.

I. Theory Checks and PI Selection

At 1 kW, identify $K \simeq 510.10\,\mathrm{W/V}$ from the measured return ratio. Compare $T(s) \simeq K(K_{pv}+K_{iv}/s)/(CVs+2P/V)$ within the measured 2–40 Hz range only.

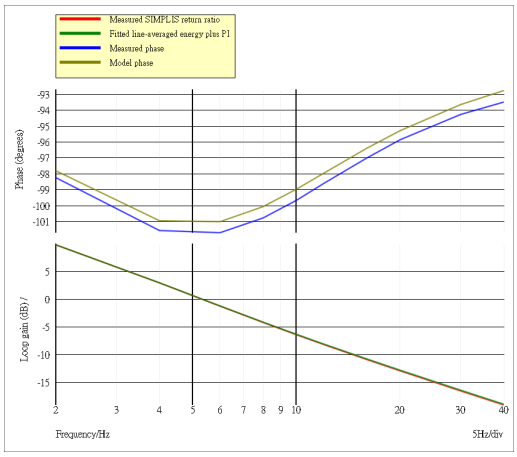


Fig. 41. Baseline return ratio versus fitted energy/PI theory. Maximum deviation: 0.148 dB and 0.723° over the measured tones.

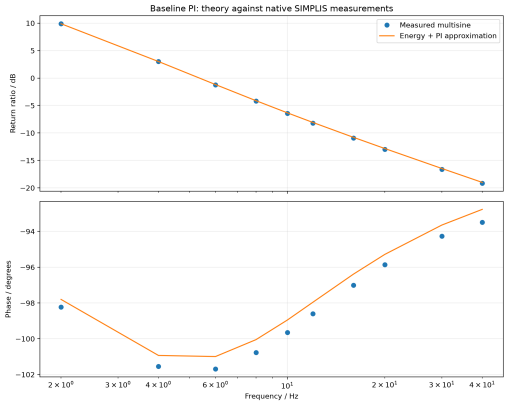


Fig. 42. Additional evidence: postprocessed SIMPLIS data. Measured return ratio and fitted model over the same frequency range.

$$\frac{\widehat{v}}{\widehat{p}_\ell} = -\frac{s}{CVs^2 + (2P/V + KK_{pv})s + KK_{iv}}. \tag{5.56}$$

Linearize at 500 W; use $K = 514.31\,\mathrm{W/V}$ from steady amplifier-output endpoints. Apply the 50 W up/down steps and the same centered 10 ms filter to theory and simulation; no timing fit.

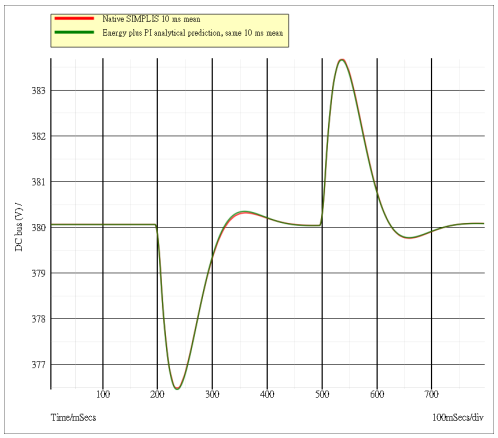


Fig. 43. 500 ↔ 550 W local theory check. Filtered maximum error: 0.0438 V; RMSE: 0.0195 V.

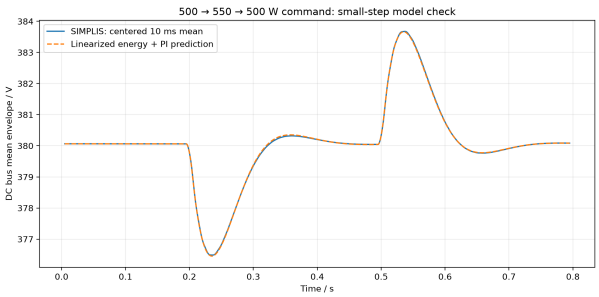


Fig. 44. Additional evidence: postprocessed SIMPLIS data. Local energy/PI prediction versus SIMPLIS for the small load step.

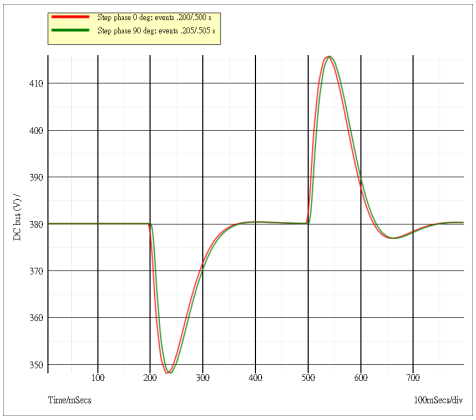


Fig. 45. Baseline step envelopes at 0° and 90° line phases; these two cases give similar recovery.

TABLE XXI
BASELINE LINE-PHASE CHECK

Phase / °	$v_{\min}$ / V	$v_{\max}$ / V	$t_{\uparrow}$ / ms	$t_{\downarrow}$ / ms
0	340.45	420.17	120.0	110.1
90	340.68	420.23	119.9	110.0

Adjacent-window probe changes are $\leq$ 0.142%; native/compact transient-envelope mismatch is < 0.020 V. Native extrema govern excursion values. Other line phases and bandwidth stability at other loads remain outside this sweep.

TABLE XXII
BASELINE SELECTION AND SCOPE

Criterion	Baseline result / interpretation
Selected PI	$\lambda = 1$, $R_z = 5.6\,\mathrm{k\Omega}$, $C_z = 4.7\,\mu\mathrm{F}$; $f_c = 5.34$ Hz, PM $= 78.35°$.
Current quality	$\mathrm{THD}_{2:40} = 3.714\%$, PF $= 0.99516$; below the 5% comparison target.
DC regulation	380.057 V mean; +0.0151% relative to 380 V.
Large steps	340.45–420.17 V raw bus; 120.0/110.2 ms recovery.
Tradeoff	Faster PI: smaller excursion and faster recovery; higher current THD.
Scope	Selected compromise; acceptance depends on actual bus and harmonic limits.

Tighter bus limits would require reassessing stored energy or the control architecture. Increasing ordinary PI bandwidth alone raises current distortion. These results do not establish hardware efficiency, harmonic compliance, cold-start behavior, or an exact UCC28180 implementation [6], [7].

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